MIL-I-27680C(AV)
9 FEBRUARY 1973
Superseding
MIL-I-27680B(AV)
31 March 1967

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, ARU-12/A, REMOTE

1. SCOPE

1.1 This specification covers a remote attitude indicator, designated ARU-12/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)

* MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

STANDARDS

FEDERAL

Fed. Std. No. 595	Colors

* MILITARY

MIL-STD-129	Marking and Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimiliar Metals
MS25237	Lamp, Incandescent, Single Contact Midget Flanged Base (T-1-3/4 Bulb)
MS33545	Case--Instrument, 5 X 5-1/4, Standard Dimensions for
MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form of

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the currenversion before use.

3. REQUIREMENTS

3.1 Qualification.--Indicator furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

* 3.2 Design characteristics of indicator.--The indicator design shall be as shown on figures 1, 2, 3, and 4. It shall consist of: two servo mechanisms for actuating the pitch and roll gimbals, a dial display, pitch trim knob, roll trim knob, external zero adjustments for pitch and roll, power-failure indicating device, and instrument case. The indicator shall be designed and constructed so that the complete mechanism can be removed from the case as a unit, and adjustments and repairs can easily be made by personnel of overhaul facilities. The design shall allow gyro input signals to be applied and removed repeatedly, with no damage to the indicator, while power is continuously applied. It shall withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Weight of the indicator shall not exceed 5-1/2 pounds.

3.3 Materials.--Materials shall be as specified herein; unless otherwise specified, materials shall be selected as specified in MIL-STD-143.

3.3.1 Metals.--Metals shall be of the corrosion-resistant type, or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions liable to be met in storage or normal service.

3.3.2 Nonmagnetic.--Nonmagnetic materials shall be used for all parts, except where magnetic materials are essential.

3.3.3 Fungusproof.--Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used, and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity; however, if used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.4 Corrosive fumes.--Materials as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4 Construction.--Construction shall be as shown on figures 1, 2, 3, and 4.

3.4.1 Dimensions and tolerances.--Unless dimensions and tolerances are specified, they shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability of parts, operation, or performance of the indicator, they shall be held or limited accordingly. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

* 3.4.2 Dissimilar metals.--Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Protective treatments.--When materials used are subject to deterioration when exposed to climatic and environmental conditions liable to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance of the indicator. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.4 Threaded parts.--Unless otherwise specified, machine screws 0.060-inch or larger in diameter shall conform to MIL-S-7742.

3.4.5 Soldering.--Soldering shall be in accordance with MIL-S-6872.

3.4.6 Riveting.--Rivets shall be tight and satisfactorily headed.

3.4.7 Screw assemblies.--Screw assemblies and bolts shall be tight. "Tight" means that the screw or bolt cannot be appreciably tightened without damage or injury to the screw or bolt.

3.4.8 Finishes and protective coatings.--

3.4.8.1 Anodizing.--Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

(a) Dials, small holes, and case inserts.

(b) Parts in a hermetically sealed container.

(c) When a part is plated with tin over a copper flash.

3.4.8.2 Chemical film.--Coating with a chemical film conforming to MIL-C-5541 may be used instead of anodizing when:

(a) Primary purpose of the treatment is to afford a suitable paint base.

(b) Castings contain nonaluminum alloy integral inserts.

* (c) Aluminum alloys that do not anodize satisfactorily are used.

3.4.8.3 Abrasion resistance.--If abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 may not be used instead of anodizing.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.

3 4.8.4 Cadmium plating.--Cadmium plating shall be in accordance with QQ-P-416, type II or III, as applicable and of a class adequate to achieve the degree of protection required.

3.4.9 Dial display markings.--Unless otherwise specified, all numerals and letters on the indicator dial display components shall be in accordance with MS33558. All markings on parts or assemblies shall be as shown on figures 1, 2, and 3. All visible portions shall be in color No. 37038 of Fed. Std. No. 595. Markings shall be clear and sharp. Dimensions shall be as specified and, on any given mark, the width shall not vary noticeably throughout the length of the mark. All markings which, for some given indication, become adjacent and which are specified to have the same basic dimension at the adjacent point, shall not be noticeably different at that point.

3.5 Servo mechanisms.--The indicator shall be provided with two servo mechanisms for actuating the pitch and roll gimbals.

3.6 Dial display.--The dial display shall be in accordance with figure 1. It shall consist basically of a fixed minature airplane, a bank angle dial, bank index, a moving two-colored spheroid background with a distinct horizon line dividing the two colors, and a mechanism mask. Unless otherwise specified, all colors shall be in accordance with Fed. Std. No. 595. The paint shall be such that the light will be sufficiently diffused to avoid highlights when the paint is illuminated with external or internal light sources. Markings shall be clear and sharp when viewed under such conditions.

3.6.1 Miniature airplane.--The minature airplane shall be in accordance with figure 2. The rear edge of the miniature airplane shall be within 1/16 inch of the spheroid surface. The wings shall follow the curvature of the spheroid. The top edge of the wings and the center dot shall be tilted back 45 degrees to facilitate night lighting. The miniature airplane shall be black, color No. 37038. The W need not have a black border as shown in figure 2; however, the white portion of the W, wings, and dot shall be separated by an area of color No. 37038 of sufficient width to show contrast. The white area shall be color No. 37875.

3.6.2 Bank angle dial.--The bank angle dial shall be in accordance with figure 2 and shall be as close to the wedge as practicable. Scale markings of the dial shall be color No. 37875.

3.6.3 Bank index.--The bank index shall be as shown in figure 2. It shall be affixed to, and rotate with, the roll gimbal but shall not move with pitch deviations. The front surface of the bank index shall be in a

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the currentversion before use.

plane parallel to the cover glass. This plane shall be in the same plane as the front surface of the small diameter of the bank dial within plus or minus 3/64 inch. The arc described by the small end of the bank index shall not deviate more than 3/64 inch radially from the small diameter portion of the bank dial. The index shall be color No. 37875.

3.6.4 Mechanism mask.--A mask, with the visible surface shaped as the surface of the frustrum of a cone, shall be attached to the roll gimbal, shall extend from the bank angle dial, and shall be approximately 3-11/16 inches on the large diameter. The outer edge of the mask shall be hidden by the inner diameter of the bank angle dial. The small diameter of the mask shall be closest to the spheroid and shall intersect the spheroid in a manner to provide a lateral opening of at least 3-3/8 inches. The inner edge of the mask shall conform to the shape of the spheroid to within 1/16 inch, plus or minus 1/32 inch.

3.6.5 Spheroid.--The spheroid shall be in accordance with figure 3. It shall be mounted in the roll gimbal, and shall rotate with the gimbal about the roll axis for roll indications, and in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator. The pitch axis shall be horizontal when the indicator is level and indicating a zero roll attitude. The horizon line on the spheroid shall be fixed to the spheroid at the zero pitch position. The spheroid shall be driven in pitch so that during a dive maneuver, the horizon line will go above the miniature airplane. The surface of the spheroid shall move 0.034 inch (nominal) for each degree of pitch change. The spheroid shall be internally illuminated. Colors and markings shall be in accordance with figure 3, and the following:

3.6.5.1 Colors and markings.--Above the horizon line, the spheroid shall be light gray, color No. 36373, and all markings shall be color No. 37038. The horizon line shall be colors No. 37038 and No. 37875, in alternating spaces, as shown on figure 3. Below the horizon line, the spheroid shall be color No. 37038, and all the markings color No. 36440. Lengths of the major pitch lines shall be as follows:

Degrees	Lengths of lines
0 (horizon line)	Width of sphere
10	2-1/2 inches
20	2-3/8 inches
30	2-1/4 inches
40	2-1/8 inches
50	2 inches
60	1-7/8 inches
70	1-3/4 inches
80	Circle
85	Solid dot.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the current version before use.

The center of the solid dots shall be plus or minus 90-degrees pitch position, and from center to center shall represent 180 degrees of pitch indication. The pitch indication shall be linear over the full range and shall be spaced over 180 degrees of the spheriod surface.

* 3.7 Pitch trim knob.--Means shall be incorporated to adjust the position of the spheroid with a pitch trim knob mounted on the lower right-hand side of the indicator and marked as shown on figure 1. Knob shall be gray, color No. 36231 of Fed. Std. No. 595. Reference mark on the knob shall be black, color No. 37038 and shall extend across the end of the knob and continue along the side to a point on the knob that is adjacent to the triangular mark on the case at the zero pitch trim position. The triangular mark shall be white, color No. 37875. The knob shall not exceed one-half turn in either clockwise (cw) or counter-clockwise (ccw) direction from the zero pitch trim position. The freedom in each direction shall be limited by a position stop. When the knob is rotated in the cw direction, the spheroid shall rotate to deflect the horizon line upward to indicate a minimum 20-degree dive. When the knob is rotated in the ccw direction, the spheriod shall rotate to deflect the horizon line downward to indicate a minimum 20-degree climb. The servo system (see 3.5) shall operate properly for any pitch trim adjustment. The knob shall have sufficient friction to remain fixed when set, but shall move smoothly without evidence of binding or erratic motion when being adjusted. The pitch trim circuit shall be designed so that if three ARU-12/A indicators are connected to a single transmitter (pitch and roll), varying the pitch trim on any one of the indicators will not affect the other indicators by more than 1/4 degree in the pitch and roll axes.

* 3.8 Roll trim knob.--The position of the spheroid shall be adjustable by means of a roll trim knob mounted on the upper right-hand side of the indicator face and marked as shown on figure 1. The reference mark on the knob shall be color No. 37038; it shall extend across the end of the knob and continue along the side to a point on the knob that is adjacent to the fulcrum mark on the case at the zero roll trim position. This mark shall be color No. 37875. The knob shall be color No. 36231. The knob shall position the bank index (3.6.3) 8 to 20 degrees right bank when rotated to its extreme cw position, and 8 to 20 degrees left bank when rotated to its extreme ccw position.

3.9 Zeroing adjustments.--With an input signal equal to plus or minus 3-degree deflection of the test transmitters (see 4.6.5) in both pitch and roll axes, the indicator shall meet the pitch and roll trim requirements of 3.7 and 3.8. When the indicator is delivered, the initial setting of the pitch and roll indication shall indicate zero pitch and roll within plus or minus 1/2 degree when the indicator is connected to a test transmitter set at electrical zero. The pitch and roll adjustments shall be combined with the pitch and roll trim circuits by means of settable knobs on the indicator face.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:42Z
Check the source to verify that this is the current version before use.

* 3.10 Power failure indicating device.--A power-failure warning flag shall be incorporated in the lower left of the face of the indicator on the bank angle dial, as shown on figure 2. The word OFF shall be black on a white background. A lead shall be brought out through the electrical connector through pin J and the circuit shall be designed so that when this pin is connected to pin A and power is applied, the warning shall disappear. When the connection or the power is removed, or both, the warning shall appear. Insofar as practicable, the warning shall also appear due to electrical failure within the indicator. The OFF letters shall be illuminated under night lighting conditions. The lighting shall not be entirely dependent upon one light source.

* 3.11 Instrument case.--The case shall be hermetically sealed. The case shall be as shown in MS33545 and in accordance with figure 4; where there are differences, figure 4 shall govern. Case shall be black, color No. 37038.

3.11.1 Cover glass.--Cover glass thickness shall not exceed 0.200 inch. Glass shall be flat and free from discoloration, scratches, and striae that interfere with normal indicator reading.

3.11.2 Filling medium.--The filling medium shall be of at least 98-percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point minus 65° C.) at the filling pressure. The filling medium shall be 100-percent helium, or a mixture of 88- to 92-percent nitrogen and the remainder helium, or 95-percent helium and 5-percent carbon dioxide. The absolute pressure of the filling medium in the case shall be approximately 1 or 1/2 atmosphere, as required. Where practicable, the 100-percent helium filling shall be used.

3.11.3 Leak rate.--The case shall be sealed so that the maximum leak rate will not permit more than 10-percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 1 atmosphere.

3.12 Lighting system.--An integral lighting system shall be provided. Supplemental lighting for the spheroid may be obtained by transillumination from within the spheroid. The integral lighting shall be in accordance with MIL-L-25467, except that the voltage shall be 28, plus or minus 1 volt instead of 5, plus or minus 1 volt. However, the gray portion of the spheroid shall at no time exceed 60 percent of the minimum brightness of the white portions of the miniature airplane.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

3.12.1 Pins.--Pins K and L shall be used for lighting purposes. The lighting leads, or lead, shall not be ground to the indicator case.

* 3.12.2 Resistors.--It is preferable that resistors do not form any part of the lighting system; however, available space shall be allocated for incorporating a dimming resistor. In cases where resistors are approved, they shall be in series with the entire lighting circuit.

3.12.3 Prime standard.--Using parts that are identical to the parts to be used in production indicators, an indicator shall be constructed which meets all display requirements of MIL-L-25467. Except for parts that affect the lighting design, this indicator may be a lighted non-operating model. The indicator model shall be tested to the above-mentioned display requirements under the surveillance of a Government inspector who will certify its conformance with MIL-L-25467 and its use as a prime standard.

3.12.4 Prime standard life.--A minimum of 12 average areas, covering all representative areas of the face of the prime standard shall be measured for brightness and color prior to submittal, and at least every 50 hours of operation thereafter. A change of plus or minus 20 percent in brightness, or 25 percent in color, in any two or more of these areas, using the initial data as reference, shall void certification of the prime standard.

3.13 Lamp. A lamp conforming to MS25237 shall be used as the light source.

3.14 Electronic components.--The indicator and its electronic components shall be in accordance with MIL-E-5400. The electrical connector shall serve all signals, power, and lighting circuits, and shall mate with type MRE 14S-G connector of the Winchester Electronics Division of Litton Industries, Inc., or equivalent. It shall be equipped with a cap or plug and gasket in a manner to protect the electrical surfaces from water vapor. The pin connections shall be in accordance with table III. The mating connector shall be secured to the indicator with a quick-disconnect mechanism.

* 3.15 Electromagnetic-interference suppression.--The indicator shall be equipped for the suppression of electromagnetic interference in accordance with MIL-STD-461.

3.16 Performance.--The indicator shall perform satisfactorily as specified herein:

3.16.1 Attitude actuation.--The spheroid (see 3.6.5) shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the control transformer synchros on

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

the indicator axes and the transmitter synchros in the pitch and roll displacement reference. The freedom of the spheroid rotation shall be unlimited about the pitch and roll axes. The gear design shall eliminate play and backlash. There shall be no hunting or jumping movement in the indicator when it is properly connected to the vertical gyro control. Overshooting shall be held to a minimum and shall not exceed one overswing with a maximum amplitude of 3 degrees with a step input equal to the maximum follow-up lag. The indicator shall not lag the signal source by more than 10 degrees at maximum rates of servo operation.

3.16.1.1 Synchro control transformer.--The synchro control transformer on the pitch and roll axes shall be high-temperature synchros, type AY 500-5 of Bendix Corporation, Eclipse-Pioneer Division, or equivalent. When four indicators are connected to a single test transmitter per axis, there shall be no noticeable change in performance under the following conditions:

(a) When the nominal no-load output of a transmitter drops .72 volt.

(b) When the phase angle between the transmitter rotor excitation voltage and the stator leg-to-leg voltage changes by 3.5 degrees.

3.16.1.2 Standard test transmitter.--The standard test transmitter shall consist of a high-precision synchro with minimum and known errors, type AY 200-1 of the Bendix Corporation, Eclipse-Pioneer Division, or equivalent. The rotor of the synchro shall be settable and rotatable through 360 degrees. These settings shall be readable to within 6 minutes of arc. Each degree of scale marking shall represent 1 degree of synchro rotor rotation with a scale marked from 0 to 360 degrees and affixed to the synchro as to be readable, as specified herein.

3.16.1.2.1 Calibration.--Method used to calibrate the standard test transmitter shall be as follows: The rotor leads shall be designated as H and C, and one stator lead designated as Z and connected to C. When 26 volts are applied across H and C (C being ground), the scale shall be positioned on zero when the voltage across H and X is maximum and the voltage across the remaining two stator leads (X and Y) is minimum. The voltage across X and Z increases before it decreases for increasing numbers of the scale.

3.16.1.2.2 Synchro excitation.--The test transmitter shall be excited from the same power supply voltage that operates the indicator. It shall be applied so that the voltage from rotor lead C to rotor lead H is in phase with the voltage from pin A to pin B of the indicator connector with lead C to pin A common.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the currenversion before use.

3.16.1.3 Pitch and roll sensing.--When the standard test transmitters, set at a scale reading of 180 degrees, are connected to the indicator, as shown in tables I and II, the indication shall drive to zero pitch and roll. Increasing scale setting of the test transmitters shall result in right bank and climb indications. Decreasing scale setting shall result in left bank and dive indications.

TABLE I. -- Roll servo connections.

Test transmitter terminal	Indicator terminal of power input connector	Motor excitation of standard test transmitter
Hot (H) Cold (C)		26-volt tapoff from single phase power supply to pins A and B of indicator
X Y Z	E D F	

3.16.2 Follow-up operation.--The electrical and mechanical design of the indicator shall be such that in the process of follow-up, the indications on pitch and bank will move smoothly without hunting, oscillating, or jumping more than 1/4 degree total amplitude in pitch or total amplitude in bank when connected to a standard test transmitter. There shall be no sustained electrical or mechanical oscillations in either axis exceeding 1/4 degree total amplitude at any indication.

3.16.2.1 Rate.--The rate at which the motors drive the gimbals shall be not less than 300 degrees per second in roll, nor 90 degrees per second in pitch.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.

TABLE II. -- Pitch servo connections.

Test transmitter terminal	Indicator terminal of power input connector	Motor excitation of standard test transmitter
Hot (H) Cold (C)		26-volt tapoff from single phase power supply to pins A and B of indicator
X Y Z	H P R	

TABLE III. -- Power input connections.

Electrical pins	Connections
A	Neutral (grounded to case).
B	115 volts, 400 cps, single-phase power in phase with excitation to test transmitter synchros.
C	Pitch null test terminal (pitch error voltage between this pin and pin A).
D, E, F	The three stator leads from the roll test transmitter synchro.
R, H, P	The three stator leads from the pitch test transmitter.
J	Power failure warning (connect to pin A to complete circuit).
K, L	Internal lighting power input.
M	No connection.
N	Roll null test terminal (roll error voltage between this pin and pin A).

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

3.16.2.2 Accuracy.--The pitch and roll indicator shall follow the input signals from their respective transmitter synchro to within plus or minus 1 degree for positions plus or minus 30 degrees in roll and pitch and 2 degrees for all other positions.

3.16.3 Amplifier channels.--The amplifier shall be an integral part of the indicator. There shall be a minimum of three stages in the transistorized amplifier with redundant transistor stages. The indicator shall continue to operate with reduced performance should partial failure of the amplifier occur.

3.16.4 Electrical power system.--The power supplied to the indicator shall be 115 volts, single-phase, 400 cycles per second (cps), in accordance with MIL-STD-704. Power input connections shall be in accordance with table III.

3.16.4.1 Electrical power variation.--The indicator shall operate properly and continuously for any alternating current (ac) voltage between 103 and 117 volts, and for any frequency between 380 and 420 cps, or any combination thereof, in addition to transient variations specified in MIL-STD-704.

* 3.16.4.2 Input signal variations.--With power continuously applied, the indicator shall operate without damage while repeated gyro input signals are applied and removed. The signal application intervals shall be one minute on and five minutes off.

3.16.5 Environmental.--The indicator shall perform satisfactorily when subjected to the following conditions, as specified in 4.6.21:

(a) Operating temperatures ranging from minus 54° C. to plus 71° C. (see 4.6.21.2 and 4.6.21.3) and a storage temperature of minus 62° C.

(b) Relative humidity up to 100 percent (see 4.6.21.7).

(c) Pressures ranging from 57 inches Hg down to 0.82-inch Hg, approximately 80,000 feet altitude (see 4.6.21.5).

(d) Exposure in salt sea atmosphere (see 4.6.21.10).

(e) A potential of 500 volts direct current (dc) for a 10-second period.

(f) Vibration incident to service use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

(g) Fungus growth as encountered in tropical climates.

(h) Sand and dust particles as encountered in desert areas.

(i) Rainfall as encountered in any locale.

(j) Acceleration forces as anticipated.

(k) Immersion in water with pressure applied, or leak test conducted with an approved helium leak detector, without evidence of leakage exceeding that specified in 3.10.3.

3.16.6 Mean-time-between-failure (MTBF).--The specified MTBF shall not be less than 1,000 hours when tested in accordance with 4.6.24.

3.17 Lubrication.--The indicator shall be lubricated in conformance with MIL-STD-838.

3.18 Identification of product.--The indicator shall be identified with a nameplate in accordance with MIL-STD-130.

3.19 Workmanship.--The indicator, including all parts and assemblies, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, tightness of assembly screws and bolts. The indicator shall be thoroughly cleaned, and loose, spattered or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests.--Examinations and test of the indicator shall be classified as (1) qualification and (2) conformance.

4.3 Test conditions.--

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

4.3.1 Standard atmospheric.--Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92-inches Hg) and at room temperature (approximately plus 25° C). When tests are made with atmospheric pressure or room temperature differing materially from these values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Voltage input.--Unless otherwise specified, the indicator shall be tested with 115 volt, 400 cps (nominal) single-phase power.

4.3.3 Properly connected.--When the indicator is referred to as being properly connected, it shall be understood that it is connected to a test fixture that is providing a lighting circuit voltage variable from 28 volt (ac or dc) to approximately 0 volt, synchro signals in accordance with 3.16.1 through 3.16.1.3 ac power as required, and with pin J jumped to pin A through a closed switch.

4.3.4 Visual examination.--Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of 2 feet.

4.4 Qualification testing.--Qualification tests of indicator shall consist of all tests specified herein. Test samples shall consist of a minimum of three indicators representative of the production equipment. Samples shall be identified with the part number and such other information as required by the procuring activity (see 6.3).

4.5 Conformance tests.--Conformance tests shall consist of (1) individual testing and (2) sampling testing.

4.5.1 Individual.--Each indicator shall be subjected to the following tests as specified in 4.6:

(a) Examination of product.
(b) Starting.
(c) Pitch and roll trim.
(d) Zeroing adjustments.
(e) Sensitivity.
(f) Follow-up rate.
(g) Follow-up accuracy.
(h) Gimbal freedom.
(i) Leak rate.
(j) Power warning indicator.
(k) Fogging.
(l) Lighting.
(m) Pitch and bank zero.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

4.5.2 Sampling plans and tests.--

* 4.5.2.1 Plan A.--One indicator shall be selected at random from each 100 or fewer produced on the contract or order and subjected to the following tests as specified in 4.6:

(a) Individual tests.
(b) Power consumption.
(c) Power variation.
(d) Follow-up operation (except overshooting).
(e) Magnetic effect.
(f) Acceleration (with power applied).
(g) Dielectric strength.
(h) Vibration error.
(i) Pressurization.
(j) Low-temperature operation.
(k) High-temperature operation.
(l) Input signal variation

* 4.5.2.2 Plan B.--Unless otherwise specified (see 6.2), three indicators selected at random from the first 15 items on the contract or order shall be subjected to the following tests as specified in 4.6:

(a) Sampling plan A tests.
(b) Electromagnetic interference.
(c) High-temperature exposure.
(d) High altitude - low temperature.
(e) Temperature shock.
(f) Humidity.
(g) Fungus.
(h) Rain.
(i) Salt spray.
(j) Sand and dust.
(k) Acceleration (without power).
(l) Reliability.
(m) Overshooting.
(n) Vibration failure.

NOTE: At the option of the procuring activity, the tests may be divided into three groups as follows, to be conducted on three sets of three instruments each, if a need exists to expedite tests. The exact number of instruments used in Group III tests is the option of the contractor; however, a minimum of three shall be used. Each indicator shall meet the requirements of all the individual tests after completing the group of specific tests to which submitted. At the option of the procuring activity, group II may be conducted on empty cases which have all external parts installed or attached as on a completed indicator. (See 6.2).

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

Group I	Group II	Group III
* Electromagnetic interference	Humidity	Vibration failure
High-temperature exposure	Fungus	Acceleration
Low temperature	Rain	Reliability tests
Temperature shock	Salt spray	Overshooting
	Sand and dust	

4.5.2.3 Rejection and retest.--When one item selected from the product run fails to meet the tests, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue.--For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that items comply with this specification.

4.5.3 Defects in items already accepted.--The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.--

4.6.1 Examination of product.--The indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

4.6.2 Starting.--The indicator shall be properly connected and power applied. The warning flag shall disappear and the indicator shall be ready for immediate operation.

4.6.3 Pitch and bank zero.--The standard test transmitters in the pitch and roll axes shall be set in the electrical zero position. Power shall be applied and the indicator shall follow up to the zero indications in pitch and bank within 1/2-degree from any random indication in either axis. This test shall be repeated and the indication shall return to within 1/4 degree of the original indication.

4.6.4 Pitch and roll trim.--

* 4.6.4.1 Pitch trim.--The pitch trim knob shall be adjusted to its cw and ccw limits and the deflection of the horizon line observed. The deflection shall be a minimum of 20 degrees from the level flight indication in both climb and dive positions. With the pitch trim knob in the zero pitch trim position, the indicator shall indicate zero

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:42Z
Check the source to verify that this is the current version before use.

pitch within one-half of the horizon line width. The knob shall have sufficient friction to remain fixed when set, but shall move smoothly without binding or erratic motion when being adjusted. The torque required to adjust the pitch trim knob shall be determined and shall be between 2- and 6-ounce inches. Two ARU-12/A indicators shall be connected to the same transmitter and the pitch trim knob moved through its range of motion on one of the indicators. There shall be no more than 1/2-degree change in the indication of the other indicator.

4.6.4.2 Roll trim.--The roll trim adjustment shall position the bank index 8 to 20 degrees right bank when the roll trim knob is rotated to its extreme cw position and 8 to 20 degrees left bank when turned to its extreme ccw position. The torque required to adjust the roll trim knob shall be determined and shall be between 2- and 6-ounce inches.

4.6.5 Zeroing adjustments.--The indicator shall be connected to the standard test transmitters, with each transmitter set to electrical zero. The pitch test transmitter shall be rotated 3 degrees in climb, the pitch trim knob loosened on its shaft, the pitch trim shaft adjusted to indicate zero, and the pitch trim knob tightened again to the shaft in the zero position. Except for the torque determination for adjusting the roll trim knob, the requirements of 4.6.4.1 shall be met. This test shall be repeated for a 3-degree dive setting on the pitch transmitter. The roll test transmitter shall be rotated 3 degrees in right roll, the roll trim knob loosened on its shaft, the roll trim shaft adjusted to indicate zero, and the roll trim knob tightened again to the shaft in the zero position. Except for the torque determination for adjusting the roll trim knob, the requirements of 4.6.4.2 shall be met. This test shall be repeated for a 3-degree left roll setting on the test transmitter. Where sampling plan B tests specify repetition of the individual tests, this test shall not be repeated.

4.6.6 Sensitivity.--The test transmitter for the roll axis shall be moved 1/4 degree in right bank and returned to the original position and then 1/4 degree in the left roll and returned to the original position. The test transmitter for the pitch axis shall be moved 1/4 degree in dive and returned to the original position and then 1/4 degree in climb and returned to the original position. There shall be a smooth, perceptible movement of the indicator bank index and horizon line for each movement of the appropriate test transmitter. This test shall be repeated in left roll, right roll, climb, and dive with the pitch trim in each of its extreme positions, except the test transmitter shall be moved 1/2 degree in each direction.

4.6.7 Follow-up rate.--The indicator shall be properly connected to a remote signal source or standard test transmitter. The follow-up rates in pitch and roll shall be determined. The computed rate in the

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.

roll axis shall be not less than 300 degrees per second and the computed rate in the pitch axis shall be not less than 90 degrees per second.

4.6.8 Follow-up accuracy.--The rotor of the standard test transmitter in the roll axis shall be displaced to at least five random positions other than the zero position. The indicator shall position itself with the rotor of the standard test transmitter within plus or minus 1 degree for each position between 0 and 30 degrees left and right bank and within 2 degrees for any position over the remaining 300 degrees. The rotor of the standard test transmitter in the pitch axis shall be displaced to at least five random positions other than the zero position. The indicator shall position itself with the rotor of the standard test transmitter within plus or minus 1 degree for each position between 0 and 30 degrees dive or climb and within 2 degrees for any position from 30 to 90 degrees in dive or climb.

4.6.9 Gimbal freedom.--The indicator shall be properly connected and the remote signal source or standard test transmitter moved 360 degrees in roll and pitch. The indicator shall follow smoothly without sticking and overshooting, or without hunting or jumping in excess of the values specified in 4.6.16.2.

4.6.10 Leak rate.--The instrument case shall be tested for leaks by means of a mass spectrometer type of helium leak detector or other means approved by the procuring activity. The maximum leak rate shall not exceed that specified in 3.11.3.

4.6.11 Power warning indicator.--The indicator shall be properly connected, except that no power shall be applied. Power shall then be applied and the flag shall disappear. The power shall be disconnected and the flag shall immediately appear. Power shall be reapplied and the jumper from pin J to A shall be disconnected; the flag shall appear.

4.6.12. Fogging.--The indicator shall be energized and in a plus 71° C., plus or minus 2° C., controlled ambient for a minimum 1-hour period. After this period and while at this temperature, an ice cube shall be rubbed on the indicator glass face for 1 minute, plus or minus 5 seconds. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.13 Lighting.--Lighting tests shall be conducted in accordance with MIL-L-25467, except as follows. Particular attention shall be given to individual segments of the indicator when making comparisons as specified.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

* 4.6.13.1 A voltage of 28 volts, plus or minus 1 volt shall be applied to both the prime standard and the indicator being tested. The two indicators shall be visually compared and shall be identical in color.

4.6.13.2 A voltage of 15.0, plus or minus 5 volts, shall be applied to both the prime standard and the indicator to be tested. There shall be no visual difference in brightness between the indicators.

4.6.13.3 Any indicator that does not meet the tests specified above shall not be approved until it has been subjected to all display tests specified in MIL-L-25467 and sufficient data have been recorded to substantiate conformance to the tests.

4.6.14 Power consumption.--The power consumption of the indicator shall be measured with suitable meters. The ac consumption shall not exceed 20 volt-amperes. The power factor shall be maintained between 0.70 lag and 1.00.

4.6.15 Power variations.--With ac power applied, the indicator shall be properly connected and subjected to the transient power conditions of MIL-STD-704. There shall be no damage to the indicator. The indicator shall then be subjected to the tests specified in 4.6.6, except the movements shall be 1 degree in roll and 3/4-degree in pitch (4.6.3), except the bank zero movement shall be 3/4-degree (4.6.9 and 4.6.11) at each of the following voltage and frequency settings:

a. 103 volts - 380 cps.
b. 127 volts - 380 cps.
c. 103 volts - 420 cps.
d. 127 volts - 420 cps.

* 4.6.15.1 Input signal variation.--This test shall be performed with the power continuously on. It shall consist of at least twenty repetitive cycles without damage to the indicator. A cycle shall consist of:

a. Both pitch and roll trim knobs in the extreme clockwise positions.
b. Gyro input signals applied for one minute and then removed for five minutes.
c. Both pitch and roll trim knobs in the extreme counterclockwise positions.
d. Gyro input signals applied for one minute and then removed for five minutes.

4.6.16 Follow-up operation.--Suitable test equipment shall be connected to the appropriate test pins of the indicator to observe and record any hunting, jumping, lagging, or overshooting of the indication when the

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.

indicator is made to follow the standard transmitter at various constant rates in both the pitch and roll axes. This shall be possible with the indicator sealed. Data shall be monitored during the initial individual tests, preceding all other tests, and during the final individual tests following the completion of all tests. While performing the individual tests, hunting, jumping and oscillation shall not exceed plus or minus 1/8 degree. Pen recorders shall not be used to determine follow-up rates and oscillation.

4.6.16.1 Indication lag.--In the pitch axis, the indication shall not lag behind the pitch remote input transmitter by more than 5 degrees for a shaft speed of 90 degrees per second. In the roll axis, the indication shall not lag behind the roll remote input transmitter more than 5 degrees for a shaft speed of 90 degrees per second, and not more than 10 degrees for a shaft speed of 300 degrees per second.

4.6.16.2 Hunting and jumping.--There shall be no noticeable hunting and jumping of the indicator display in pitch or roll with no rate inputs from the test transmitters. The test transmitter shafts shall be rotated separately through all rates from zero through the maximum speed of the respective axis of display. The survey of rates shall be sufficiently slow to record the hunting and jumping that occurs at any speed of each axis of display of the indicator and shall take not less than 1 minute with the rate varying approximately linearly. The peak-to-peak amplitude of the hunting and jumping shall not exceed 1/8 degree for shaft speeds between 0 and 20 degrees per second in pitch and roll, 1/2 degree for shaft speeds between 20 and 90 degrees per second in pitch and roll, nor 1 degree for shaft speeds between 90 and 300 degrees per second in roll.

4.6.16.3 Overshooting.--A step input equal to the maximum allowable follow-up lag shall be induced in each axis of the indicator. Overshooting shall not exceed one overswing with a maximum amplitude of 3 degrees.

4.6.17 Magnetic property.--The magnetic property test shall be conducted first with no power applied to the indicator and then with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of, and 5 inches from the center of, a free magnet approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 oersted, plus or minus 0.01 oersted. The maximum deflection of the magnet shall not exceed 5 degrees. An aircraft compass with the compensating magnets removed may be used as the free magnet.

4.6.18 Acceleration (operating).--The attitude shall be set at zero pitch and roll with each axis displaced 90 degrees. The indicator shall be properly connected and power applied to the attitude circuit. The

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the currentversion before use.

indicator shall be accelerated along its normal vertical axis with a force of 5g for 60 seconds; the attitude display shall not vary from the initial indication by more than 1 degree in pitch or roll.

4.6.19 Dielectric.--A potential of 500 volts dc shall be applied between isolated pins and between pins and the case for a 10-second period. There shall be no breakdown of insulation.

4.6.20 Pressurization.--The indicators shall be tested in a chamber at 57 inches Hg, plus or minus 1 inch, and 25° C. for a 4-hour period. At the end of this period, and while still at this pressure and temperature, the indicator shall be subjected to the test specified in 4.6.9. At the conclusion of this test, the indicator shall be closely examined for evidence of deformation. Any deformation, damage, or unsatisfactory operation shall be cause for rejection. After this test, the indicator shall be subjected to, and pass, the leak rate test.

4.6.21 Environmental tests.--The following environmental tests shall be conducted.

4.6.21.1 Vibration error.--The indicator shall be mounted as specified in procedure IV of MIL-E-5272, properly connected, and subjected to vibration with a total amplitude between 0.003 and 0.005 inch.

4.6.21.1.1 Roll and pitch oscillation.--The maximum total amplitude of roll and pitch indicated oscillation throughout the frequency range shall be noted and shall not exceed 1/4 degree.

4.6.21.1.2 Roll and pitch variation.--The maximum variation of the roll and pitch indication during vibration from the original indication before vibration, throughout the frequency range shall be noted and shall not exceed 1 degree.

4.6.21.2 Low-temperature operation.--The indicator shall be properly connected, except that no power shall be applied, and placed in a chamber at a temperature of minus 54° C., plus or minus 2° C., for a 4-hour period. At the end of this period and while the indicator is at the reduced temperature, power shall be applied for a 5-minute period. While still at the reduced temperature, the indicator shall then be subjected to, and meet, the tests specified in 4.6.3, except the tolerance shall be plus or minus 1 degree, and 4.6.6, except the tolerance in pitch and roll with the pitch trim knob in the zero position shall be plus or minus 1/2 degree, and the tolerance in pitch and roll with the trim knobs in each of their extreme positions shall be plus or minus 1 degree.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

4.6.21.3 High-temperature operation.--The indicator shall be properly connected, placed in a chamber at a temperature of 71° C., plus or minus 2° C., and with power applied for a 4-hour period. At the end of this period and while at the high temperature, the indicator shall meet the tests specified in 4.6.3, except that tolerance shall be plus or minus 1 degree: and 4.6.6, except the zero pitch trim tolerance shall be plus of minus 1/2 degree and plus or minus 1 degree with the pitch trim at the extremes.

4.6.21.4 High-temperature exposure.--The indicator shall be subjected to a high-temperature exposure test in accordance with procedure II of MIL-E-5272, except the equipment shall be exposed for 24 hours and no power shall be applied. At the conclusion of the 24-hour period and while at the high temperature, the indicator shall be subjected to, and shall pass, the test specified in 4.6.9.

4.6.21.5 High altitude - low temperature.--A high altitude - low temperature test shall be made in accordance with procedure II of MIL-E-5272. The chamber shall be reduced to minus 54° C., with the indicator properly connected but no power applied. After a period of 27 hours, the pressure in the chamber shall be reduced to the equivalent of 80,000 feet, plus or minus 500 feet, and power applied. The time required for the indicator to start operating shall not exceed 5 minutes. There shall be no binding or sticking at any indication. After the indicator has returned to room temperature and the pressure has been increased to atmospheric. the indicator shall meet all individual tests.

4.6.21.6 Temperature shock.--The temperature shock test shall be conducted in accordance with procedure I of MIL-F-5272. At the conclusion of this test, the indicator shall be subjected to, and shall pass, the individual tests.

4.6.21.7 Humidity.--The indicator shall be subjected to a humidity test in accordance with procedure I of MIL-E-5272. There shall be no sticking or binding in the pitch trim knob, nor evidence of corrosion or rust of the internal and external parts of the indicator as a result of this test.

4.6.21.8 Fungus.--The fungus test shall be conducted in accordance with procedure I of MIL-E-5272. There shall be no sticking or binding in the pitch trim knob and fungus growth shall not be supported.

4.6.21.9 Rain.--The rain test shall be conducted in accordance with procedure II of MIL-E-5272, except that the indicator shall be mounted for 1/2 hour with the dial in a vertical plane and 1/2 hour with the

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the current version before use.

dial in a horizontal plane (dial face upward). Upon removal from the chamber, no sticking or binding shall occur in the pitch trim knob as a result of this test.

4.6.21.10 Salt spray.--The salt spray test shall be conducted in accordance with procedure I of MIL-E-5272. At the end of the 50-hour period, there shall be no sticking or binding of the pitch trim knob, nor damage to the indicator as a result of this test. The tolerance specified for the individual tests shall apply.

4.6.21.11 Sand and dust.--The sand and dust test shall be conducted in accordance with procedure I of MIL-E-5272. At the end of this test, the indicator shall be removed and allowed to cool at room temperature. There shall be no sticking or binding of the pitch trim knob.

* 4.6.21.12 Vibration failure.--A vibration failure test shall be conducted in accordance with vibration test Part I, curve III of MIL-T-5422. In addition, the indicator shall be vibrated for 50 hours on each of its three axes at 11.5 cycles per second and 0.13 inch double amplitude (0.8g). The indicator shall be operating during all these vibration tests. At the conclusion of this test, the indicator shall meet the test specified in 4.6.3. No looseness in the mechanism, nor damage to any part of the indicator, shall result from vibration.

4.6.22 Acceleration (nonoperating).--The indicator, not operating, shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. It shall then be subjected to an acceleration of 5g in each of these axes for a 1-minute period. At the end of this acceleration, the indicator shall meet the individual tests. No damage to the indicator shall result from this test.

* 4.6.23 Electromagnetic interference.--The electromagnetic-interference test shall be conducted in accordance with MIL-STD-461 for equipment class 1D.

4.6.24 Reliability testing.--

* 4.6.24.1 MTBF.--The specified MTBF of 1,000 hours shall be demonstrated utilizing the procedures of MIL-STD-781. The specified mean-time-between-failures (θ_0) shall be demonstrated with a discrimination ration ($\frac{\theta_0}{\theta_1}$) of 2, a consumer's decision risk (β) of 0.10, and a producer's decision risk (α) of 0.10. The sample size shall be at the option of the contractor, but shall not be less than three units. The indicators used for this test shall have successfully passed the vibration test of 4.6.21.12 and the acceleration test of 4.6.22. Instrument operating time for these tests may be credited to MTBF test time provided no failure occurred during these two tests.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the current version before use.

4.6.24.2 Test.--The indicator shall be properly connected and subjected to a cycle consisting of a 4-hour period of operation followed by a 5-minute period with no power applied. During the operating, both axes of the attitude display shall be oscillated through a minimum of plus or minus 30 degrees at a rate of 5 to 7 times per minute, the lighting circuit shall be energized with 28 volts, and the power warning flag system shall show that the indicator is operating. During the resting or cooling-off period, all power shall be removed from the instrument under test, including the power to the lighting and flag circuits. The standard test transmitter shall remain excited, and its rotor shall continue to be displaced through a minimum of plus of minus 30 degrees, at a rate of 5 to 7 times per minute. Throughout each 4-hour period of operation the test indicator shall be subjected to vibration along its vertical axis (parallel to front face) with the frequency of the applied vibration cycled at a logarithmic rate between 10 and 200 cps with a vibratory acceleration level of 0.1g. The frequency range shall be cycled from minimum to maximum in 7.5 minutes. At least once during each 4-hour period, the pitch and roll trim knobs shall be rotated to their extremes 10 times in each direction. This cycle shall be repeated throughout the reliability test.

4.6.24.3 Monitoring.--Throughout the reliability test the performance of the test indicators shall be monitored. At a period not to exceed each 24 hours of operation of each indicator under test, the indicator performance parameters specified below shall be measured and recorded. If the value of any specified indicator performance parameter is not within tolerance, a failure shall be presumed to have occurred immediately after the last successful measurement of the same parameter.

a. Pitch and bank zero (4.6.3).
b. Pitch and roll trim (4.6.4).
c. Sensitivity (4.6.6).
d. Follow-up rate (4.6.7).
e. Follow-up accuracy (4.6.8).
f. Gimbal freedom (4.6.9).
g. Power warning indicator (4.6.11).
h. Follow-up operation (4.6.16).
i. Power consumption (4.6.14).
j. Lighting (4.6.13).

4.6.24.4 Tolerance.--For the purpose of the reliability test, an out-of-tolerance condition shall be as follows.

4.6.24.4.1 Where a tolerance-type requirement is specified, a failure shall be considered as any electrical or mechanical conditions which exceeds the specified tolerance by 100 percent.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.

4.6.24.4.2 Where a maximum, or minimum, type requirement is specified, a failure shall be considered as any electrical or mechanical condition which results in a value greater than the specified maximum, or less than the specified minimum by 5 percent of the specified value.

4.6.24.4.3 Where a condition is specified as a requirement, a failure shall be considered as any electrical or mechanical malfunction which results in any condition other than that specified.

4.6.24.5 Test length.--The total unit hours of test time and unit hours each indicator shall be monitored continuously. It is intended that no indicator shall accumulate less time than the lesser of:

a. The specified MTBF.
b. One-half of the average operating time of all indicators on test.

If any individual indicator fails to meet the above-mentioned requirement, every effort shall be made to repair the equipment and return it to test. The test shall continue until the total unit hours of test, together with the total count of relevant equipment failures permit either an accept or a reject decision in accordance with the specified accept-reject criteria. All references to "test time" or "test duration" are to be understood as meaning indicator "on" time, or indicator operating time for purposes of MTBF determination.

4.7 Inspection of preparation for delivery.--Preservation and packaging, packing, and marking shall be inspected to insure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.--Preservation and packaging shall be level A, in accordance with method III of MIL-P-116.

5.2 Packing.--

★ 5.2.1 Level A or B.--Inciators preserved and packaged as specified in 5.1, shall be packed for shipment in accordance with level A, or level B, of MIL-STD-794, as specified in 6.2.

★ 5.2.2 Air shipment.--Indicators shall be packed in accordance with level B of MIL-STD-794.

5.3 Marking for shipment.--In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129. Packing nomenclature shall be INDICATOR, ATTITUDE, ARU-12/A, REMOTE.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.--The ARU-12/A remote attitude indicator is intended for use in helicopters. The indicator will be controlled by a vertical reference control that will provide transmitting synchro pitch and roll output signals.

6.2 Ordering data.--Procurement data should specify the following:

a. Title, number, and date of this specification.
b. Whether sampling plan B tests are to be conducted; whether tests may be grouped; and whether group II tests may be conducted on empty cases (see 4.5.2.2).
c. Level of packing required (see 5.2.1).
d. Whether special marking for shipment is requirement (see 5.3).

* 6.3 Qualification.--With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Products proposed for listing on the Qualified Products Listing shall be fully identified by manufacturer's part number and shall be so listed when qualified. (see 4.4). The activity responsible for the Qualified Products List is US Army Aviation Systems Command, ATTN: Standardization Division, RD&E, PO Box 209, Main Office, St. Louis, Missouri 63166. Information pertaining to qualification of products may be obtained from that activity.

* 6.4 DELETED

* 6.5 International standardization.--Certain provisions of this specification are the subject of international standardization agreements (ASCC Air Stds 10/6B, Numeral and Letter for Aircrew Station Displays; 10/42, Attitude Indicators and STANAGs 3319, Dimensions for Instruments Cases; 3329 Numerals and Letters; 3344 Standard Synchro Electrical Characteristics for Aircraft Instruments and 3637, Attitude Indicators.) When amendment, revision, or cancellation of this specification is proposed which affects or violates an international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the currenversion before use.

*6.6 Changes from previous issue.--The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army -- Aviation Materiel Command

Preparing Activity:

Army -- AV

Project No. 6610-A013

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

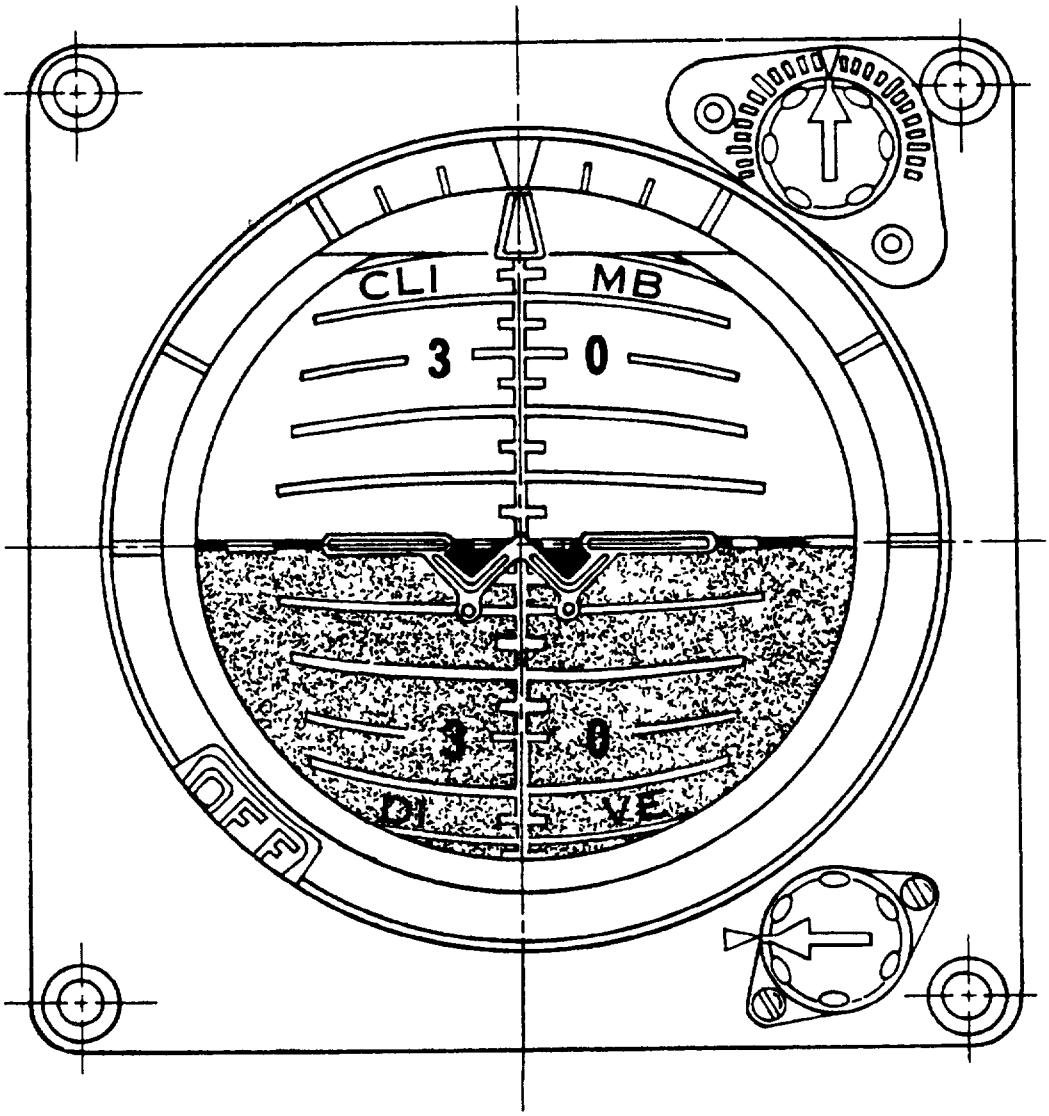
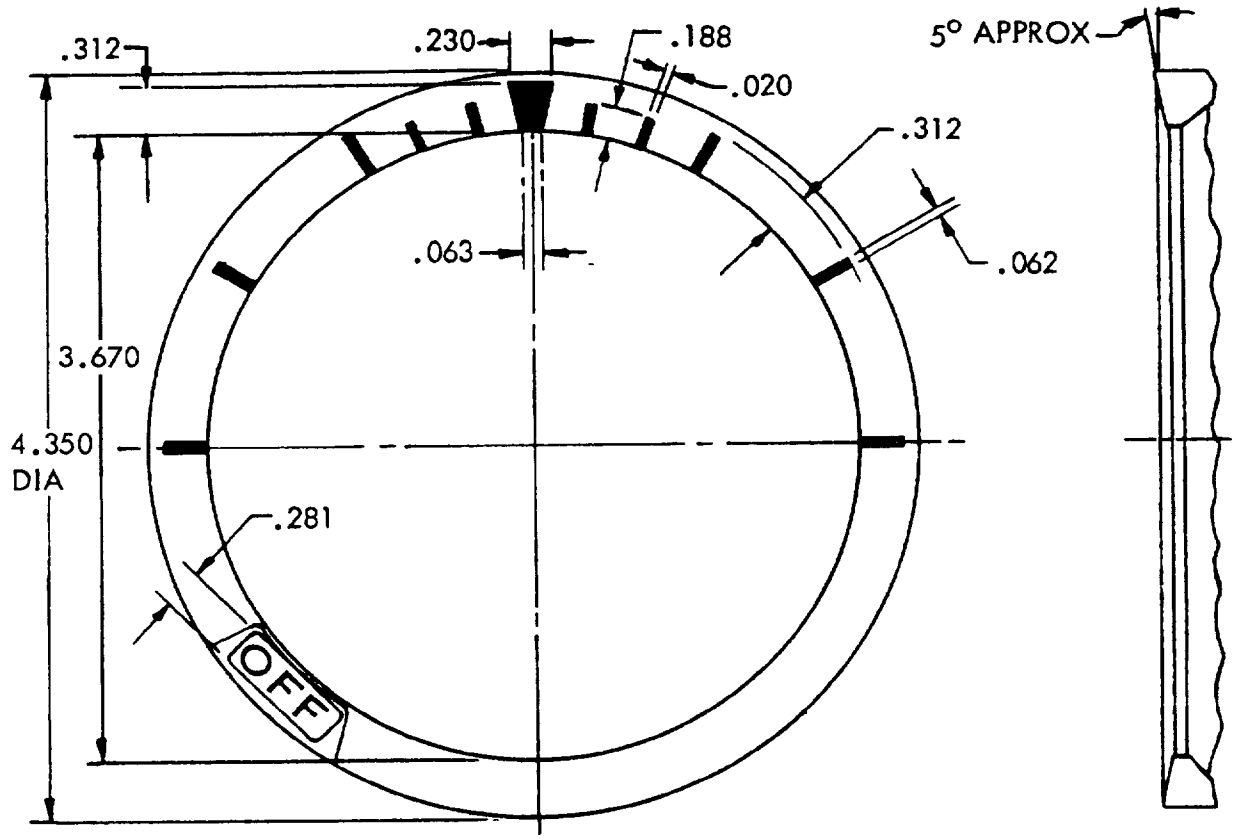
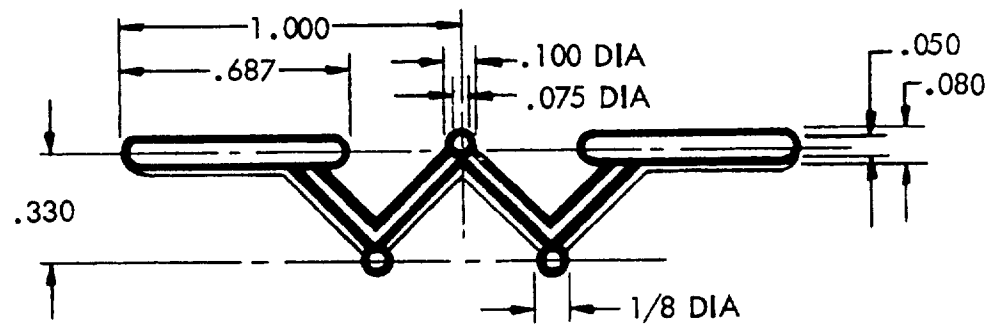


FIGURE 1. DIAL DISPLAY

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.

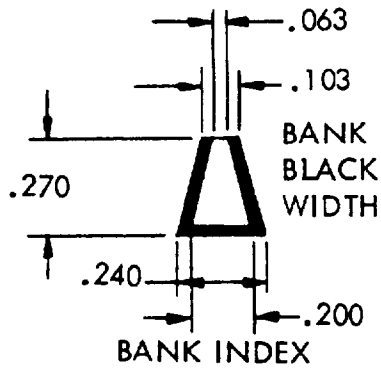


BANK SCALE



MINATURE AIRPLANE

AIRPLANE SHALL HAVE A BLACK BORDER OF UNIFORM WIDTH WITHIN .005 IN.



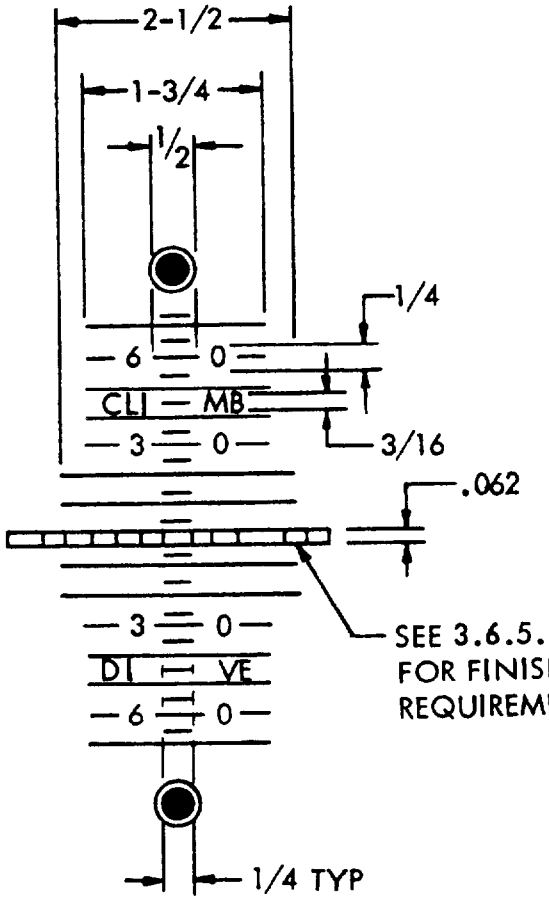
BANK INDEX

BANK INDEX SHALL HAVE A BLACK BORDER OF UNIFORM WIDTH WITHIN .005 IN

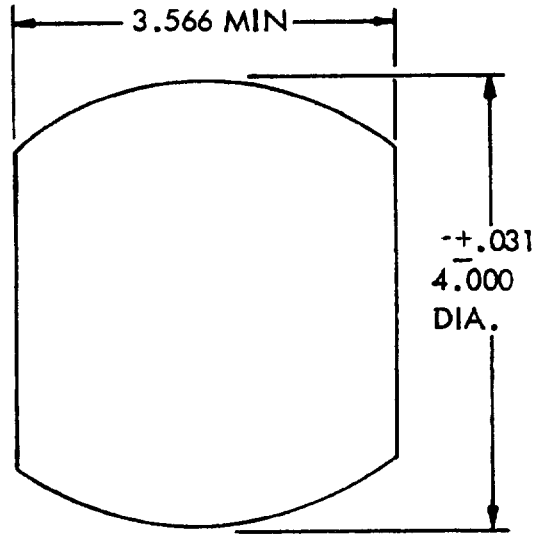
DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ± 1/64 DECIMALS ± .005,
ANGLES ± 2°

FIGURE 2. DIAL DETAILS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.



SEE 3.6.5.1
FOR FINISH
REQUIREMENTS



SPHEROID
(FRONT VIEW)

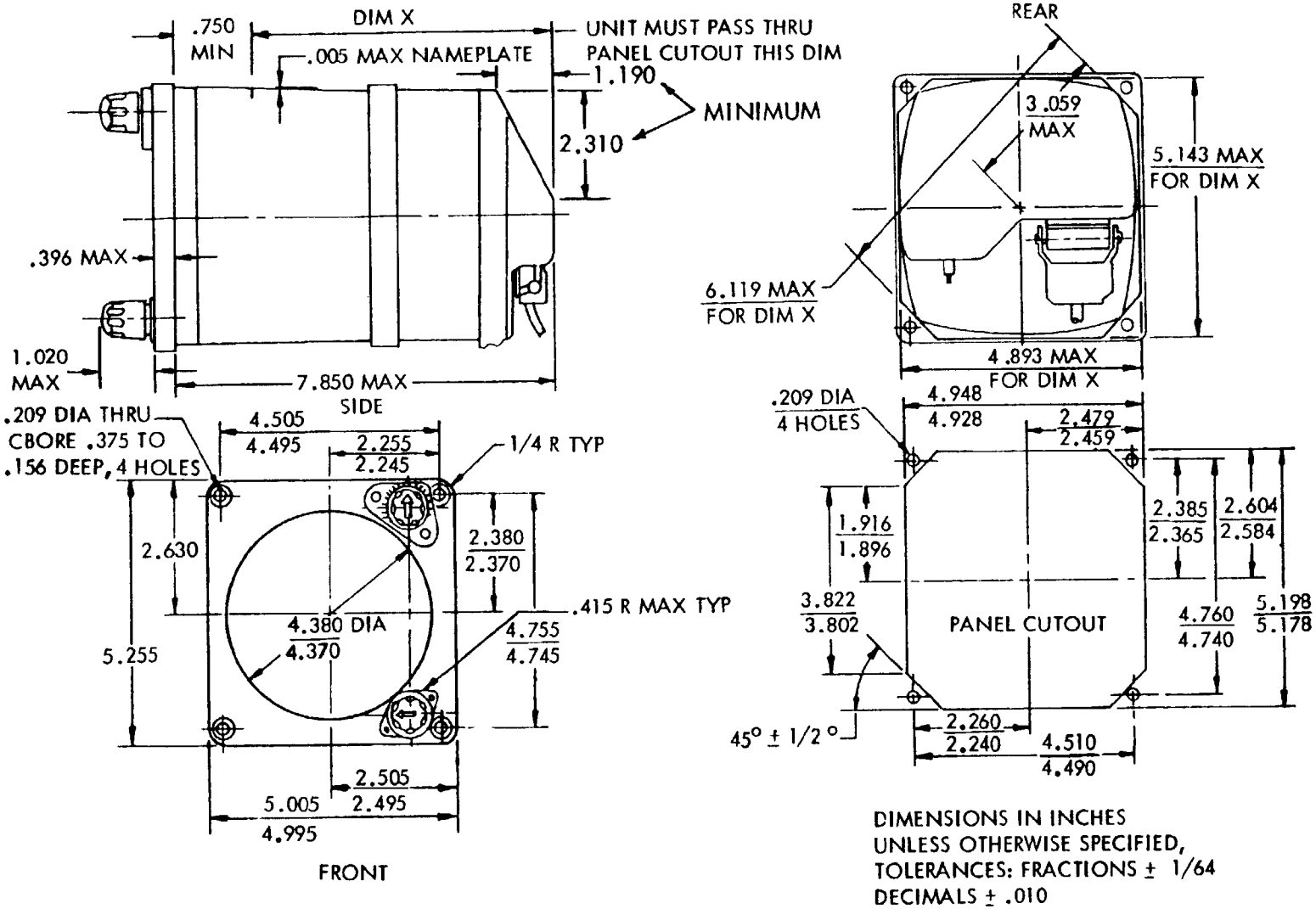
ALL MARKINGS, NUMERALS &
LETTERS SHALL BE .020 WIDE
UNLESS OTHERWISE SPECIFIED

THE PITCH REFERENCE LINES SHALL BE SO
PLACED ON THE SPHERE THAT
THEY ARE PORTIONS OF GREAT CIRCLES
WHICH INTERSECT AT THE PITCH AXIS

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES:
FRACTIONS $\pm$ 1/64, DECIMALS $\pm$ .005, ANGLES $\pm 2^{\circ}$

FIGURE 3. SPHEROID AND MARKINGS

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:42Z
Check the source to verify that this is the current version before use.



* FIGURE 4. CASE OUTLINE

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:42Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-20T12:42Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
US Army Aviation Systems Command
ATTN: AMSAV-R-MN
PO Box 209
St. Louis, Missouri 63166

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:42Z
Check the source to verify that this is the current version before use.